

Logic in Computer Science
Problem Sheet 4: Logical Equivalence
September 12, 2025

I. Prove that

(a) $(A \vee (B \wedge C)) \equiv ((A \vee B) \wedge (A \vee C))$

(b) $(\neg(A \rightarrow B)) \equiv (A \wedge (\neg B))$

(c) $(\neg(A \wedge B)) \equiv ((\neg A) \vee (\neg B))$

2. Check whether the following pairs of formulas are logically equivalent.

(a) $(\neg(p_1 \vee ((\neg p_1) \wedge p_2))), ((\neg p_1) \wedge (\neg p_2))$

(b) $((p_1 \rightarrow p_2) \rightarrow p_1), (p_1 \rightarrow (p_2 \rightarrow p_1))$

(c) $(p_1 \rightarrow (p_2 \rightarrow p_1)), (p_1 \vee (\neg p_1))$

3. Convert the following formulas into CNF and DNF.

(a) $(\neg(((\neg p) \rightarrow q) \rightarrow r))$

(b) $(p \vee q) \wedge (\neg q \vee r \vee \neg s) \wedge (r \vee \neg p)$

(c) $(p \leftrightarrow (q \vee r))$

(d) $((\neg p) \rightarrow q) \vee ((p \wedge (\neg r)) \leftrightarrow q)$

4. (a) Define recursively the set of formulas without negation operator.

(b) Prove that any formula without the negation operator is satisfiable.

5. Carry out the Tseitin encoding on the formula $(p_1 \rightarrow (p_3 \wedge p_4) \vee (p_2 \rightarrow (p_3 \wedge p_5)))$ for finding an equi-satisfiable CNF formula.

6. Prove or disprove: If B is the Tseitin encoding of a formula A then $A \equiv B$.

7. Show the following set of operators defines $\neg, \vee, \wedge, \rightarrow$ and $\leftrightarrow$.

(a) $\{\vee, \wedge, \neg\}$

(b) $\{\rightarrow, \neg\}$

(c) $\{\uparrow\}$ the set containing only nand operator

(d) $\{\downarrow\}$, the set containing only nor operator

8. Prove or disprove:

(a) $\{\wedge, \vee\}$ can define $\neg$.

(b) $\{\vee, \wedge, \rightarrow, \leftrightarrow\}$ can define $\neg$.

9. Show that $\{\wedge, \leftrightarrow, +\}$ can define $\neg, \vee, \wedge, \rightarrow$ and $\leftrightarrow$ but no proper subset can do that. Here $+$ is xor operator.